

Progress Test 2.A — Thermo-S1

2.A · Thermoelastic behaviour of condensed phases

Semester 1 · Academic Year 2026–2027

ANSWER KEY · MODEL ANSWERS AND MARKING SCHEME

Duration: 10 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Unit conversion

(1 mark)

The isothermal compressibility of water is $\chi_T = 4.5 \times 10^{-10} \text{ Pa}^{-1}$. Express it in bar^{-1} .

$1 \text{ bar} = 10^5 \text{ Pa, so } 1 \text{ Pa}^{-1} = 10^5 \text{ bar}^{-1}$	$\chi_T = 4.5 \times 10^{-10} \times 10^5 \text{ bar}^{-1}$
$\chi_T = 4.5 \times 10^{-5} \text{ bar}^{-1}$	

Marking

①
0 1
☐ ☐

Marking scheme

① 1 pt $4.5 \times 10^{-5} \text{ bar}^{-1}$.

Examiner's tip

“Per pascal” to “per bar”: a bar is 10^5 times larger than a pascal, so the coefficient per bar is 10^5 times larger. A quick sanity check: water loses 4.5×10^{-5} of its volume per bar.

Exercise 2. Thermoelastic coefficients — definitions

(5 marks)

Define the isothermal compressibility χ_T and the isobaric expansion coefficient α of a condensed phase. Give their SI units. Explain the minus sign in χ_T , and why both definitions divide by V .

$\chi_T = -\frac{1}{V} \left(\frac{\partial V}{\partial p} \right)_T$ in Pa^{-1}	$\alpha = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_p$ in K^{-1}
Minus sign: when p increases at fixed T, V decreases, so $\left(\frac{\partial V}{\partial p} \right)_T < 0$. The minus sign makes $\chi_T > 0$.	
Division by V: $\left(\frac{\partial V}{\partial p} \right)_T$ and $\left(\frac{\partial V}{\partial T} \right)_p$ are extensive. Dividing by V gives intensive coefficients, which do not depend on the size of the sample.	

Marking

①
0 1 2
☐ ☐ ☐
②
0 1 2
☐ ☐ ☐
③
0 1
☐ ☐

Marking scheme

- ① **2 pts** $\chi_T = -\frac{1}{V} \left(\frac{\partial V}{\partial p} \right)_T$, with the fixed variable T written.
 ② **2 pts** $\alpha = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_p$, with the fixed variable p written.
 ③ **1 pt** Units Pa^{-1} and K^{-1} ; sign and intensive character explained.

Exercise 3. Condensed phases

(3 marks)

For a liquid or a solid, the isothermal compressibility χ_T is:

- ☒ very small ($\chi_T \approx 0$) ☐ very large
☐ negative ☐ infinite

Why: “condensed phases are weakly compressible”: their volume hardly changes when p increases.

Expressed with the mass density ρ , χ_T equals:

- ☒ $\frac{1}{\rho} \left(\frac{\partial \rho}{\partial p} \right)_T$ ☐ $-\frac{1}{\rho} \left(\frac{\partial \rho}{\partial p} \right)_T$
☐ $\rho \left(\frac{\partial \rho}{\partial p} \right)_T$ ☐ $-\frac{1}{V} \left(\frac{\partial \rho}{\partial p} \right)_T$

Why: for a closed system $\rho = m/V$, so compressing increases ρ . The minus sign disappears (equation (7) of the notes).

Over a restricted temperature range, the coefficient α of a condensed phase can be considered:

- ☒ constant ☐ zero
☐ proportional to p ☐ negative

Why: remark (ii) of Box 11. This is what allows the double integration of Box 13.

Exercise 4. Expansion and compression of water

(5 marks)

The coefficients of water are assumed constant. (a) A volume V_0 of water is heated at constant pressure from θ_1 to θ_2 . Estimate the change of volume ΔV . (b) Water is compressed at constant temperature from p_1 to p_2 . Estimate its relative change of volume $\Delta V/V$.

$$\alpha = 2.0 \times 10^{-4} \text{ K}^{-1} \quad \chi_T = 5.0 \times 10^{-10} \text{ Pa}^{-1} \quad V_0 = 250 \text{ mL}$$

$$\theta_1 = 20^\circ\text{C} \quad \theta_2 = 70^\circ\text{C} \quad p_1 = 1 \text{ bar} \quad p_2 = 101 \text{ bar}$$

(a) At fixed p , $\frac{dV}{V} = \alpha dT$. For a small change, $\Delta V \approx \alpha V_0 \Delta T$ with $\Delta T = 70 - 20 = 50 \text{ K}$ (a difference in $^\circ\text{C}$ equals a difference in K):

$$\Delta V \approx 2.0 \times 10^{-4} \times 250 \text{ mL} \times 50 \quad \Delta V \approx 2.5 \text{ mL} \quad (1\% \text{ of } V_0)$$

(b) At fixed T , $\frac{dV}{V} = -\chi_T dp$, so $\frac{\Delta V}{V} \approx -\chi_T \Delta p$ with

$$\Delta p = 100 \text{ bar} = 1.0 \times 10^7 \text{ Pa:}$$

$$\frac{\Delta V}{V} \approx -5.0 \times 10^{-10} \times 1.0 \times 10^7 \quad \frac{\Delta V}{V} \approx -5.0 \times 10^{-3} = -0.5\%$$

Even 100 bar changes the volume by only 0.5%: water is almost incompressible.

Marking

①	0	1
②	0	1
③	0	1
④	0	1
⑤	0	1

Marking scheme

- ① 1 pt $\Delta V \approx \alpha V_0 \Delta T$ (literal).
- ② 1 pt $\Delta T = 50 \text{ K}$ and $\Delta V \approx 2.5 \text{ mL}$.
- ③ 1 pt $\Delta V/V \approx -\chi_T \Delta p$ (literal, with the minus sign).
- ④ 1 pt $\Delta p = 1.0 \times 10^7 \text{ Pa}$.
- ⑤ 1 pt $\Delta V/V \approx -5.0 \times 10^{-3}$, interpreted.

Examiner's tip

A *difference* of temperature has the same value in K and in $^\circ\text{C}$, so here $\Delta T = 50 \text{ K}$. Only absolute temperatures (as in $pV = nRT$) must be converted.

Your score: / 14 $\longrightarrow$ mark out of 20 = score $\times$ 20/14 = / 20